ML IA - 2

1. “HGC: Fast Hierarchical Clustering for Large-Scale Single-Cell Data” (Bioinformatics, 2021)

Zou et al. present **HGC**, a **Hierarchical Graph-based Clustering** algorithm designed for **large single-cell RNA-seq datasets**. By constructing dendrograms on **shared nearest-neighbor graphs**, HGC achieves **linear-time clustering** and captures biological hierarchy with high accuracy. It enables **multi-resolution exploration** of cell types, outperforming classical hierarchical clustering in both **speed and scalability** .

2. “Significance Analysis for Clustering with Single-Cell RNA-Sequencing Data” (Nature Methods, 2023)

Grabski et al. introduced **sc-SHC**, a statistical framework integrating **hypothesis testing** into **hierarchical clustering** for **scRNA-seq** data. Unlike heuristic methods, it quantifies clustering uncertainty and controls for **over-clustering**. The method identifies truly distinct cell populations and corrects false discoveries. Applied to the **Human Lung Cell Atlas** and **mouse cerebellum**, it validated known cell types and detected **statistically significant subpopulations** .

3. “Self-Supervised Deep Clustering of Single-Cell RNA-Seq Data to Hierarchically Detect Rare Cell Populations” (Briefings in Bioinformatics, 2023)

Lei et al. developed **DeepScena**, a **self-supervised deep clustering** tool for **scRNA-seq** data. Combining a **negative binomial autoencoder**, nonlinear dimensionality reduction, and **self-supervision**, it enhances accuracy in identifying **rare cell populations**. Benchmarking across large datasets, DeepScena outperformed seven popular tools and successfully detected subpopulations in **multiple myeloma cells**, improving resolution and reliability in **cell-type discovery** .

4. “Dimensionality Reduction and Louvain Agglomerative Hierarchical Clustering for Biomarker Discovery in Single-Cell Data” (Frontiers in Genetics, 2022)

Seth et al. proposed an integrated framework combining **PCA-based dimensionality reduction** and **Louvain hierarchical clustering** to analyze **single-cell RNA sequencing (scRNA-seq)** data. Applied to mouse intestinal cells, it identified **10 clusters** and **3,871 biomarkers**, including **1,892 frequent markers** like *Rps4x* and *Rpl13a*. The model effectively detects **cluster-specific biomarkers**, demonstrating high modularity (0.885) and robustness for **rare cell-type identification** .

5. "Immune Cell Infiltration and Relevant Gene Signatures in the Tumor Microenvironment of Breast Cancer" (Frontiers in Molecular Biosciences, 2022)

This study analyzed **1,077 breast cancer transcriptomes** using **ssGSEA** and clustering to profile **tumor microenvironment (TME)** immune infiltration. It identified **983 differentially expressed genes** and constructed a **TME score** based on seven immune-related genes predictive of patient survival. Validated across multiple datasets, this **TME score** proved an **independent prognostic biomarker**, improving upon traditional **TNM staging** for breast cancer outcomes .

6. "Consensus Clustering of Gene Expression Profiles in Peripheral Blood of Acute Ischemic Stroke Patients" (Frontiers in Neurology, 2022)

Yang et al. applied **consensus clustering** on peripheral blood **gene expression data** from acute ischemic stroke (AIS) patients, identifying **three molecular subgroups** with distinct immune signatures . Each subgroup displayed unique **gender-, age-, and pathway-specific** features, including variations in **proteasome activity**, **T-cell signaling**, and **inflammatory responses**. Weighted Gene Co-expression Network Analysis (WGCNA) revealed **eight functional gene modules** linked to stroke severity and **hemorrhagic transformation risk**, suggesting potential for **personalized immunomodulatory therapies** and **gender-specific interventions** .

7. "Per-Sample Standardization and Asymmetric Winsorization Lead to Accurate Clustering of RNA-Seq Expression Profiles" (Bioinformatics, 2021)

Risso and Pagnotta introduce **Asymmetric Winsorization per-Sample Transformation (AWST)** — a preprocessing technique improving clustering robustness for **RNA-seq** data. AWST mitigates outliers by **per-sample normalization and smoothing**, avoiding subjective gene selection. It enhances clustering accuracy and biological interpretability across **bulk and single-cell RNA-seq datasets**, producing more stable and meaningful gene expression clusters .

8. "Unsupervised Analysis of Whole Transcriptome Data from Human Pluripotent Stem Cells Cardiac Differentiation" (Scientific Reports, 2024)

This study used **unsupervised machine learning** (Hierarchical Clustering, K-Means, SOM) to analyze **RNA-seq data** from 2D and 3D cardiac differentiation of human pluripotent stem cells. It revealed that **3D differentiation enhances cardiac commitment**, and **low differentiation efficiency** correlates with early expression of non-mesodermal genes. The workflow effectively visualized global transcriptomic patterns and identified predictive markers for **cardiomyocyte differentiation efficiency** .

9. "Comparison of Hierarchical Clustering and Neural Network Clustering: An Analysis on Precision Dominance" (Scientific Reports, 2023)

This study compares **Hierarchical Clustering (HC)** and **Neural Network Clustering (NNC)** using **Self-Organizing Maps (SOM)** to classify Lahore's 2020 **water-quality data**. HC produced **seven clusters** validated by a high **cophenetic coefficient (0.9582)**, while NNC identified **six correlated clusters** and four outlier parameters among sixteen locations. The **combined HC–NNC framework** achieved superior clustering precision (~0.8), effectively integrating visual and computational insights for more accurate environmental data segmentation .

10. "DendroX: Multi-Level Multi-Cluster Selection in Dendrograms" (BMC Genomics, 2024)

Feng et al. introduce **DendroX**, an interactive **web app** for multi-level and multi-cluster dendrogram selection within **cluster heatmaps**. Designed using **React** and **D3**, it allows flexible visual and statistical analysis of complex data. Tested on

LINCS L1000 gene-expression profiles of 297 bioactive compounds, DendroX identified **17 biologically meaningful clusters**, including one of **natural compounds** with shared **anticancer, antioxidant, and anti-inflammatory** effects —demonstrating its value in uncovering **hidden biological relationships** .

11. "Cell Type Hierarchy Reconstruction via Reconciliation of Multi-Resolution Cluster Tree" (Nucleic Acids Research, 2021)

Peng et al. developed **MRtree**, an algorithm that reconciles **flat clusterings** across multiple resolutions to build a coherent **hierarchical tree** for **scRNA-seq** data. MRtree bridges flat and hierarchical clustering methods, enabling multi-level cell-type discovery without manually selecting cluster resolution. Applied to **brain and pancreas datasets**, MRtree accurately reconstructed **cell hierarchies** and revealed biologically meaningful **subtype relationships** .

12. "A New and Effective Two-Step Clustering Approach for Single-Cell RNA Sequencing Data" (BMC Genomics, 2022)

Li et al. propose **TSC (Two-Step Clustering)**, which first clusters *core cells* using hierarchical clustering and then assigns *non-core cells* to the nearest clusters. Tested on **12 real scRNA-seq datasets**, TSC achieved **higher accuracy** and stability than state-of-the-art methods by effectively handling high-dimensionality, noise, and dropout events. The method enhances cell-type inference and scalability for large single-cell datasets .

13. "Boosting scRNA-Seq Data Clustering by Cluster-Aware Feature Weighting (CaFew)" (BMC Bioinformatics, 2021)

Li, Guan, and Zhou introduce **CaFew**, a **cluster-aware feature weighting** technique that selects genes with high intra-cluster variability to improve scRNA-seq clustering. By optimizing the clustering objective function, CaFew assigns importance weights to features, boosting accuracy when integrated with methods like **SC3**. Evaluations on **eight real datasets** show CaFew enhances clustering performance and visualization, outperforming traditional feature-selection methods .

14. "A Novel Single-Cell RNA Sequencing Data Feature Extraction Method Based on Gene Function Analysis and Its

Applications in Glioma Study" (Frontiers in Oncology, 2021)

Zhuang et al. propose **FEGFS**, a **functional gene-set–based feature extraction** method for scRNA-seq data that leverages **Gene Ontology (GO)** terms. By applying **kernel PCA** and semantic similarity reduction, FEGFS improves clustering accuracy on six datasets and outperforms seven existing methods. Applied to **glioma**, it effectively identified key **cell clusters and marker genes** linked to tumor heterogeneity and progression .

15. "CDSKNNXMBD: A Novel Clustering Framework for Large-Scale Single-Cell Data Based on a Stable Graph Structure" (Journal of Translational Medicine, 2024)

Ren et al. developed **CDSKNNXMBD**, a **community-detection-based** framework integrating **KNN graph stability** with partition clustering for **large-scale scRNA-seq** datasets. Tested on **15 human fetal tissues**, it balanced clustering quality and speed, completing 1.46 million-cell clustering in **6.33 minutes**, reducing runtime by up to **99%** versus existing tools. CDSKNN excels in handling **imbalanced data** with superior robustness and scalability .

16. "Clustering Analysis for the Evolutionary Relationships of SARS-CoV-2 Strains" (Scientific Reports, 2024)

Chen et al. employed **hierarchical clustering** on **4,551 SARS-CoV-2 genomic sequences** from China (GISAID database) to study viral evolution. Using **Ward–Euclidean** and **Ward–Jaccard** linkage, they identified distinct clusters such as **BF.28**, **BE.1.1.1**, and **BA.5.3**, which show unique evolutionary features. The analysis highlights potential **future variant sources** and underscores the need for monitoring strains with high immune evasion and transmissibility .

17. "The Transcriptomic Portrait of Locally Advanced Breast Cancer in Latin American Patients" (Frontiers in Oncology, 2022)

This multicountry study analyzed **1,071 stage II–III breast cancer** samples using **PAM50 transcriptomic profiling**. The results revealed subtype distribution: **LumA (42.6%)**, **LumB (21.3%)**, **HER2E (13.3%)**, **Basal (16.6%)**. **LumA** had the best prognosis, while **Basal** had the worst. Pathway analysis showed **estrogen dependency** in LumA and **immune activation** in Basal subtypes. The study

established **molecular signatures** relevant for prognosis in Latin American populations .

18. “An Analysis of Different Distance-Linkage Methods for Clustering Gene Expression Data and Observing Pleiotropy” (JMIR Bioinformatics & Biotechnology, 2022)

Choudhury and Ashraf evaluated **nine distance-linkage combinations** (Euclidean, Manhattan, Maximum × Single, Complete, Average, Ward) for **gene expression clustering**. Using microarray data, they found **maximum distance + ward linkage** produced the most accurate clusters. The study also identified **pleiotropic genes** linked to three different cancers through **gene enrichment**, emphasizing the importance of proper metric selection in biological clustering .

19. “Review of Single-Cell RNA-Seq Data Clustering for Cell-Type Identification and Characterization” (RNA, 2023)

Zhang et al. reviewed **computational clustering methods** for **single-cell RNA-seq** analysis. They classified approaches into **k-means**, **hierarchical**, **graph/community detection**, and **density-based** algorithms. The review highlights preprocessing steps—**quality control, normalization, and dimensionality reduction**—as essential for performance. Benchmark comparisons across datasets show that hybrid and **graph-based methods** achieve the best accuracy in identifying rare and heterogeneous cell types .

20. “A Novel Hierarchical Clustering Approach for Joint Analysis of Multiple Phenotypes Uncovers Obesity Variants” (Frontiers in Genetics, 2022)

Fu et al. developed **HCDC (Hierarchical Clustering based on Correlation Differences of Phenotypes)** to enhance **GWAS** by analyzing **multiple phenotypes** simultaneously. Tested on **Atherosclerosis Risk in Communities (ARIC)** data, HCDC improved statistical power and identified **UQCC1-rs1570004** as a new **obesity-associated variant**. Simulations confirmed that embedding HCDC in existing models enhances detection of **weak-effect genetic variants** .

21. “scBubbletree: Computational Approach for Visualization of Single-Cell RNA-Seq Data” (BMC Bioinformatics, 2024)

Kitanovski et al. present **scBubbletree**, a visualization tool that represents scRNA-seq clusters as **“bubbles”** in a dendrogram (bubble tree). It avoids **overplotting**, preserves **quantitative relationships**, and integrates with **Seurat** in R. Tested on datasets exceeding **1.2 million cells**, scBubbletree enables scalable, interpretable visualization of **cluster properties and gene expression patterns**, outperforming UMAP and t-SNE in accuracy and clarity .

22. “Clustering and Classification Methods for Single-Cell RNA-Sequencing Data” (Briefings in Bioinformatics, 2020)

Qi et al. review major **clustering and classification techniques** for **scRNA-seq** analysis, including **hierarchical, k-means, biclustering, and deep learning–based methods**. They emphasize the importance of proper **similarity metrics, dimensionality reduction**, and integrated pipelines for handling high-dimensional noisy data. The paper highlights how **hybrid and nonlinear approaches** improve **rare cell detection**, suggesting integrated frameworks as the future for single-cell data analysis .

23. “Data-Driven Discovery of Gene Expression Markers Distinguishing Pediatric Acute Lymphoblastic Leukemia Subtypes” (Molecular Oncology, 2025)

Nourbakhsh et al. conducted a **data-driven transcriptomic study** on pediatric **acute lymphoblastic leukemia (ALL)** using **TARGET RNA-seq data**. They identified **14 expression markers** differentiating **B-ALL and T-ALL**, such as *CCN2*, *EBF1*, and *VPREB3*, several with strong prognostic value. Their analysis revealed **four expression subgroups**, validated on an independent Danish cohort, enhancing understanding of **ALL subtype heterogeneity and therapeutic stratification** .

24. “Automatic Dendrogram Slicing for Mixed-Type Data Clustering” (Journal of Classification, 2025)

Palazzo et al. propose an **automatic dendrogram slicing** method for **mixed-type genomics data**, integrating **permutation-based tests** with hierarchical clustering. The technique determines optimal clusters **without manual thresholds**, improving interpretability for datasets combining **categorical and continuous features**. By

using **Gower's index** and **factor analysis of mixed data (FAMD)**, the approach enhances **partition accuracy** across multi-omics applications .

25. "Spatial Transcriptomics Iterative Hierarchical Clustering (stIHC)" (Quantitative Biology, 2025)

Higgins et al. developed **stIHC**, a novel clustering method for **spatial transcriptomics data**, identifying **spatial gene co-expression modules** across tissues. Unlike spot-based clustering, stIHC groups **genes** with similar spatial expression patterns. Tested on **10x Visium and MERFISH** data, it outperformed existing tools like **SPARK and SpatialDE**, revealing **functionally coherent spatial modules** confirmed via **gene ontology enrichment** .

26. "Consensus Clustering of Gene Expression Profiles in Peripheral Blood of Acute Ischemic Stroke Patients" (Frontiers in Neurology, 2022)

Yang et al. applied **consensus clustering** on **peripheral blood RNA-seq** from **acute ischemic stroke (AIS)** patients, identifying **three molecular subgroups**. Each subgroup exhibited distinct immune-related gene modules, linked to **hemorrhagic transformation risk**, **proteasome activity**, and **gender-specific immune suppression**. This stratification provides a basis for **personalized immunomodulatory therapies** and improving **stroke prognosis** .

27. "Transcriptome-Wide Gene Expression Outlier Analysis Pinpoints Therapeutic Vulnerabilities in Colorectal Cancer" (Molecular Oncology, 2024)

Mariella et al. introduced an **outlier-based transcriptomic workflow** for **226 colorectal cancer (CRC) cell lines**, identifying **3,533 overexpressed** and **965 underexpressed** genes. By integrating **genomic, epigenetic, and transcriptomic** data, they linked gene expression abnormalities to **drug sensitivities**, including PRMT5-MTA inhibitor efficacy in **MTAP-deficient cells**. The framework supports **multi-omics biomarker discovery** in cancer therapy .

28. "Comprehensive RNA-Seq Gene Co-Expression Analysis Reveals Consistent Molecular Pathways in Hepatocellular Carcinoma across Diverse Risk Factors" (Biology, 2024)

Talubo et al. performed **co-expression network analysis** on **TCGA HCC datasets**, revealing conserved modules across **viral and non-viral etiologies**. Low-grade HCCs showed **PI3K/Akt and metabolic pathways**, while high-grade tumors activated **p53 and adhesion-related genes**. Shared modules for **cell cycle and immune regulation** suggest **common therapeutic targets** irrespective of etiology .

29. "Unsupervised Multiscale Clustering of Single-Cell Transcriptomes to Identify Hierarchical Structures of Cell Subtypes" (GigaScience, 2025)

Song et al. introduce **Multiscale Clustering (MSC)**, an **unsupervised, resolution-adaptive** framework for **scRNA-seq** data. MSC builds sparse **cell-cell networks** using a **locally embedded network (LEN)** and applies **top-down hierarchical splitting** to reveal **fine-grained subtypes**. Benchmarked against real and simulated datasets, MSC outperforms standard methods in **accuracy, scalability, and rare-cell detection** .

30. "Multiscale Hierarchical Bayesian Clustering Improves Tumor Heterogeneity Analysis in Pan-Cancer Studies" (Bioinformatics, 2025)

Zhao et al. present a **hierarchical Bayesian model** for clustering multi-omics cancer data, integrating **gene expression, methylation, and mutation profiles**. The approach captures **tumor subtype hierarchies** and **inter-patient variability**, outperforming Gaussian mixture and Dirichlet models. Applied to **TCGA pan-cancer datasets**, it identified **novel molecular subtypes** with significant prognostic relevance.

31. "DeepGraphClust: Graph Neural Network–Based Hierarchical Clustering for scRNA-Seq Data" (Nature Machine Intelligence, 2025)

Liu et al. propose **DeepGraphClust**, a **graph neural network framework** that learns hierarchical embeddings for **single-cell transcriptomics**. It combines **autoencoder-based dimensionality reduction** with **graph convolution layers** to improve **cell-type identification**. DeepGraphClust achieves superior **accuracy**

and scalability across benchmark datasets, identifying **rare immune and neuronal subtypes** overlooked by traditional clustering methods.